

Q1. String Character Frequency Counter

CODE

```
str=input("enter the character\n")
count={}
for ch in str:
    if ch!=" ":
        if ch in count:
            count[ch]+=1
        else:
            count[ch]=1
for ch in sorted(count):
    print(ch,'→',count[ch])
```

INPUT hello world

OUTPUT enter the character
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
str=input("enter the sentence:")
vowels="aeiou"
vc=0
cc=0
vf={}
cf={}
for ch in str:
    if ch.isalpha():
        if ch in vowels:
            vc+=1
            if ch in vf:
                vf[ch]+=1
            else:
                vf[ch]=1
        else:
            cc+=1
            if ch in cf:
                cf[ch]+=1
            else:
                cf[ch]=1
print("vowels:",vc)
print("consonants:",cc)
if vf:
    maxvowel=max(vf,key=vf.get)
    print("most frequent vowel:",maxvowel,'(',vf[maxvowel],')')
else:
    print("most frequent vowel:none")
if cf:
    mincons=min(cf,key=cf.get)
    print("least frequent consonant:",mincons,'(',cf[mincons],')')
else:
    print("least ferquent consonant:none")
if cc!=0:
    ratio=vc/cc
    print("vowel to consonant ratio",round(ratio,2))
else:
    print('vowel to consonant ratio:undefiend')
```

INPUT hlibro

OUTPUT

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Q3. String Compression and Run-Length Encoding

CODE

```
s=input("enter the characters:")
c=""
count=1
for i in range(len(s)):
    if i<len(s)-1 and s[i]== s[i+1]:
        count+=1
    else:
        c+=s[i]
        if count>1:
            c+= str(count)
            count=1
if len(c)<len(s):
    print("\ncompressed:",c)
else:
    print("\ncompression not beneficial")
    print("original:",s)
d=""
i=0
while i< len(c):
    ch=c[i]
    i+=1
    if i<len(c) and c[i].isdigit():
        d+=ch*int(c[i])
        i+=1
    else:
        d+=ch
print("decompressed",d)
```

INPUT abc

OUTPUT enter the characters:
compression not beneficial
original: abc
decompressed abc

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Q4. Advanced Text Statistics Analyser

CODE

```

text=input("enter the text:")
words=text.split()
print("words:",len(words))
sentence=text.count(".")*text.count("!")+text.count("?")
if sentence==0:
    sentence=1
print("sentence:",sentence)
print("paragraphs: 1")
letters=0
for w in words:
    letters+=len(w)
avgwords=letters/len(words)
print("average word length:",round(avgwords,2))
avgsentence=len(words)/sentence
print("average sentence length:",round(avgsentence,2))
stop=["the", 'is', 'a', 'in', 'an', 'of', 'and', 'to', 'for']
freq={}
for w in words:
    w=w.lower().strip(".,!")
    if w not in stop:
        if w in freq:
            freq[w]+=1
        else:
            freq[w]=1
print("top 5 words:")
count=0
while count <5 and len(freq)>0:
    m=max(freq,key=freq.get)
    print(m, "(", freq[m], ") ", sep="")
    del freq[m]
    count+=1
vowels="aeiou"
syllables=0
for w in words:
    w=w.lower()
    prev=False
    for ch in w:
        if ch in vowels:
            if prev==False:
                syllables+=1
            prev= True
        else:
            prev=False
score=206.835-1.015(len(words)/sentence)-84.6*(syllables/len(words))
print("flesch reading ease:",round(score,2))

```

```
INPUT      python is a powerful programming language
```

[illegible]